INCH-POUND
MIL-DTL-32260(AR)
23 October 2007
SUPERSEDING
(See 6.9)

DETAIL SPECIFICATION

Primer, Percussion M28A2 Assembly

Inactive for new design after 23 October 2007

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers metal parts, loading, assembling and packing of the M28A2 percussion primer.

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks below form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-223	-	Powder Black
MIL-P-20449	-	primer, Percussion, M61 Loading, Assembling and Packing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition lot Numbering
MIL-STD-1916	-	DOD Preferred method of Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch, http://assist.daps.dla.mil or from the Standardization Documents Order desk, 700 Robbins Avenue, Bldg 4D. Philadelphia, PA 19111-5094.

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications below form part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

8838012	-	Cup, Battery
8838015	-	Plug, Firing
8838080	-	Head
8838089	-	Body
8838129	-	Primer, Percussion M28A2 Assembly

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from the U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM D 3951 - Standard Practice for Commercial Packaging

(Copies of this document are available online at http://www.astm.org or from ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials, components and assemblies.

3.2.1 Cup, battery. The materials, components and assemblies shall be in accordance with all requirements specified on drawing 8838012.

3.2.2 Plug, firing. The materials, components and assemblies shall be in accordance with all requirements specified on drawing 8838015.

3.2.3 Head. The materials, components and assemblies shall be in accordance with all requirements specified on drawing 8838080.

3.2.4 Body. The materials, components and assemblies shall be in accordance with all requirements specified on drawing 8838089.

3.2.5 Primer, percussion M28A2 assembly (head prior to assembling battery cup and firing plug). The materials, components and assemblies shall be in accordance with all requirements specified on drawing 8838129.

3.2.6 Primer, percussion M28A2 assembly (head prior to assembling to body). The materials, components and assemblies shall be in accordance with all requirements specified on drawing 8838129.

3.2.7 Primer, percussion M28A2 assembly (body prior to assembling head assembly). The materials, components and

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assemblies shall be in accordance with all requirements specified on drawing 8838129.

3.2.7 Primer, percussion M28A2 assembly. The materials, components and assemblies shall be in accordance with all requirements specified on drawing 8838129.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Moisture content of black powder. The moisture content of black powder at the loading station at the time of loading shall not exceed 0.5 percent.

3.5 Head Assembly.

3.5.1 Non-functioning. The head assembly shall not function by an applied energy of 47.52 inch-ounces minimum (min).

3.5.2 Functioning. The head assembly shall function by applied energy of 159.20 inch-ounces maximum (max).

3.6 Primer assembly ballistic test.

3.6.1 Functioning. The primer assembly shall function without audible hang fire.

3.6.2 Chamber pressure resistance. The primer assembly shall withstand a pressure of 54,000 to 57,000 pounds per square inch (psi) without evidence of rupture, or any part of the primer becoming detached.

3.7 Check test for deterioration of Primer, Percussion (M61 Assembly). The M61 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-20449, if the total elapsed time between original acceptance of the M61 primer lot and the assembly of that lot into the M28A2 primer exceeds two years, or if the M61 primers have been subjected to adverse conditions for any period of time since previous test.

3.8 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, skillful manner. Components shall be free of burrs, sharp edges, cracks, dirt, grease, corrosion, and other foreign matter. Cleaning methods used shall not be injurious to any parts, nor shall the cleaning agents contaminate the parts. All required marking and stamping shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement / verification cross-reference matrix.

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test	Classes of Verification A- Design Verification B- First Article C- Acceptance

Section 3 Requirements	Description	Verification Method					Verification Class			Section 4 Verification
		N/A	1	2	3	4	A	B	C	
3.1	First Article				X	X		X	X	4.2
3.2	Materials, Components and assemblies									
3.2.1	Cub, battery				X			X	X	4.3.2.1
3.2.2	Plug, firing				X			X	X	4.3.2.2
3.2.3	Head				X	X		X	X	4.3.2.3/4.4.1
3.2.4	Body				X			X	X	4.3.2.4
3.2.5	Primer, percussion M28A2 assembly				X	X		X	X	4.3.2.5-4.3.2.8/4.4.3
3.3	Ammunition lot numbering				X			X	X	4.3.1
3.4	Moisture content of black powder					X		X	X	4.4.4
3.5	Head Assembly									
3.5.1	Non-functioning					X		X	X	4.4.1
3.5.2	Functioning					X		X	X	4.4.2
3.6	Primer assembly									
3.6.1	Functioning					X		X	X	4.4.3
3.6.2	Pressure resistance					X		X	X	4.4.3
3.7	Check test for deterioration of M61 primer assembly					X			X	4.4.5
3.8	Workmanship				X			X	X	4.3.2.1-4.3.2.9

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4.1 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article verification in accordance with Table II.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.2 First article inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE II. First article inspection

PARAGRAPH	TITLE Primer, Percussion M28A2 Assembly	SHEET 1 OF 1	DRAWING NUMBER
CLASSIFICATION	EXAMINATION OR TEST	Number of FAT samples	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cup, Battery			
	Dwg. 8838012			
	Examination for Defects	25	3.2	4.3.2.1
	Plug, Firing			
	Dwg. 8838015			
	Examination for Defects	25	3.2	4.3.2.2
	Head			
	Dwg. 8838080			
	Examination for Defects	25	3.2	4.3.2.3
	Non-Functioning	15	3.5	4.4.1
	Functioning	10	3.5	4.4.2
	Body			
	Dwg. 8838089			
	Examination for Defects	25	3.2	4.3.2.4
	Primer, Percussion M28A2 Assembly			
	Dwg. 8831157			
	Examination for Defects	25	3.2	4.3.2.5 – 4.3.2.8
	Proving Ground Tests	25	3.6	4.4.3

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916 paragraph 4.2, Acceptance by Tables. In addition, each lot of primers shall contain:

a. Black Powder from not more than two lots interfix number from one manufacturer. A lot of black powder shall be used to completion.

b. Bodies and components of one lot interfix number from one manufacturer.

c. Primer, Percussion M61 (herein after referred to as primer elements) of one lot interfix number from one manufacturer.

4.3.2 Classification of characteristics

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, inspection level IV shall be used for all characteristics defined as majors and inspection level II for all minor characteristics. Critical characteristics shall be addresses in accordance with MIL-STD-1916.

b. Conformance Inspection. Conformance inspection shall be performed in accordance with paragraph 4.3.2.1 through 4.3.2.8. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1	Cup, Battery		8838012
			NEXT HIGHER ASSEMBLY 8838129

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Major				
101	Outside diameter	Level IV	3.2.1	AIE
102	Thickness through bottom	Level IV	3.2.1	AIE
103	Length	Level IV	3.2.1	AIE
104	Diameter of hole	Level IV	3.2.1	AIE
105	Inside diameter	Level IV	3.2.1	AIE
Minor				
201	Flat missing from surface at open end	Level II	3.2.1	Visual
202	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.2.2	Plug, Firing	SHEET 1 OF 1	8838015
			NEXT HIGHER ASSEMBLY 8838129

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Large diameter	Level IV	3.2.2	AIE
102	Total length	Level IV	3.2.2	AIE
103	Length of point	Level IV	3.2.2	AIE
104	Profile of point	Level IV	3.2.2	AIE
105	Length of pilot	Level IV	3.2.2	AIE
106	Diameter of pilot	Level IV	3.2.2	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		SHEET 1 OF 2	DRAWING NUMBER
4.3.2.3	Head			8838080
				NEXT HIGHER ASSEMBLY 8838129

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None, defined			
MAJOR				
101	Pitch diameter of thread	Level IV	3.2.3	AIE
102	Major diameter of thread	Level IV	3.2.3	AIE
103	Diameter of body pilot	Level IV	3.2.3	AIE
104	Diameter of battery cup cavity	Level IV	3.2.3	AIE
105	Diameter of percussion element cavity	Level IV	3.2.3	AIE
106	Diameter of under flange	Level IV	3.2.3	AIE
107	Diameter of flange	Level IV	3.2.3	AIE
108	Thickness of flange	Level IV	3.2.3	AIE
109	Depth of percussion element cavity	Level IV	3.2.3	AIE
110	Depth of battery cup cavity	Level IV	3.2.3	AIE
111	Concentricity between percussion element cavity and flash hole	Level IV	3.2.3	AIE
112	Concentricity of percussion element cavity with diameter under flange	Level IV	3.2.3	AIE
113	Concentricity of flange with diameter under flange	Level IV	3.2.3	AIE

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3 Cont.	TITLE Head	SHEET 2 OF 2	DRAWING NUMBER 8838080 / NEXT HIGHER ASSEMBLY 8838129	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MAJOR				
114	Diameter of chamfer at mouth of percussion element cavity	Level IV	3.2.3	AIE
115	Diameter of flash hole	Level IV	3.2.3	AIE
116	Length to shoulder above thread	Level IV	3.2.3	AIE
117	Length to thread undercut	Level IV	3.2.3	AIE
118	Length of thread including thread undercut	Level IV	3.2.3	AIE
		Level IV	3.2.3	AIE
119	Total length	Level IV	3.2.3	AIE
120	Depth of flash cavity	Level IV	3.2.3	AIE
121	Profile of flash cavity	Level IV	3.2.3	AIE
122	Crack or split	Level IV	3.2.3	Visual
123	Flash hole missing	Level IV	3.2.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.4	Body		8838089
			NEXT HIGHER ASSEMBLY 8838129

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Flash hole missing	100% 0,1	3.2.4	AIE
2	Tube Cracked or split	100% 0,1	3.4.4	AIE
Major				
101	Pitch diameter of thread	Level IV	3.2.4	AIE
102	Minor diameter of thread	Level IV	3.2.4	AIE
103	Length of thread	Level IV	3.2.4	AIE
104	Wall thickness	Level IV	3.2.4	AIE
105	Location of holes	Level IV	3.2.4	AIE
106	Inside diameter	Level IV	3.2.4	AIE
107	Depth of counterbore	Level IV	3.2.4	AIE
108	Diameter of counterbore	Level IV	3.2.4	AIE
109	Total length	Level IV	3.2.4	AIE
110	Concentricity of inside diameter to pitch diameter of thread	Level IV	3.2.4	AIE
Minor				
201	Protective coating damaged	Level II	3.2.4	Visual
202	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Primer, Percussion M28A2 Assembly (Head, prior to assembling battery cup and firing plug)	SHEET 1 OF 1	DRAWING NUMBER 8838129 NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None, defined			
MAJOR				
101	Percussion element missing, cocked or inverted	Level IV	3.2.5	AIE
102	Check test of primer element M61	4.4.5	3.7	MIL-P-20449
MINOR				
201	Seal between percussion element and head missing or incomplete	Level II	3.2.5	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.2.6	Primer, Percussion M28A2 Assembly (Head, prior to assembling to body)	SHEET 1 OF 1	8838129
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Firing plug or battery cup above flush with surface of head	100%	3.2.6	AIE
MAJOR				
101	Firing plug missing or inverted	Level IV	3.2.6	Visual
102	Flash hole in head completely obstructed	Level IV	3.2.6	Visual
103	Battery cup loose or missing	Level IV	3.2.6	Manual/Visual
104	Head assembly non functioning	4.4.1	3.5.1	4.4.1
105	Head assembly functioning	4.4.2	3.5.2	4.4.2
MINOR	None defined			

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.2.7	Primer, Percussion M28A2 Assembly (Body, prior to assembling head assembly)	SHEET 1 OF 1	8838129 NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None, defined			
MAJOR				
101	Weight of charge	Level IV	3.2.7	AIE
102	Liner fails to cover all holes	Level IV	3.2.7	Visual
103	Foreign matter in charge	Level IV	3.2.7	Visual
104	Moisture content of black powder	4.4.4	3.4	4.4.4
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.8	TITLE Primer, Percussion M28A2 Assembly	SHEET 1 OF 1	DRAWING NUMBER 8838129 / NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None, defined			
MAJOR				
101	Head assembly loose on body	Level IV	3.2.8	Manual
102	Any flash hole not completely sealed	Level IV	3.4.8	Visual/Note 1
103	Total length maximum	Level IV	3.2.8	AIE
104	Diameter at flange maximum	Level IV	3.2.8	AIE
105	Diameter at front end maximum	Level IV	3.2.8	AIE
106	Primer assembly functional and pressure resistance	4.4.3	3.6	4.4.3
Minor				
201	Marking missing, misleading, or unidentifiable	Level II	3.2.8	Visual
202	Crimp of body to head is missing or incomplete	Level II	3.2.8	Visual
203	Evidence of poor workmanship	Level II	3.8	Visual

Note 1. Sketches 1, 2 and 3 on Figure 1 indicate acceptable primers. Furthermore, a primer with holes displaying the conditions shown in sketches 4, 5 or 6 may be classed acceptable provided that the visible liner is thoroughly impregnated with paint or lacquer and not more than 20 percent of the holes have this condition.

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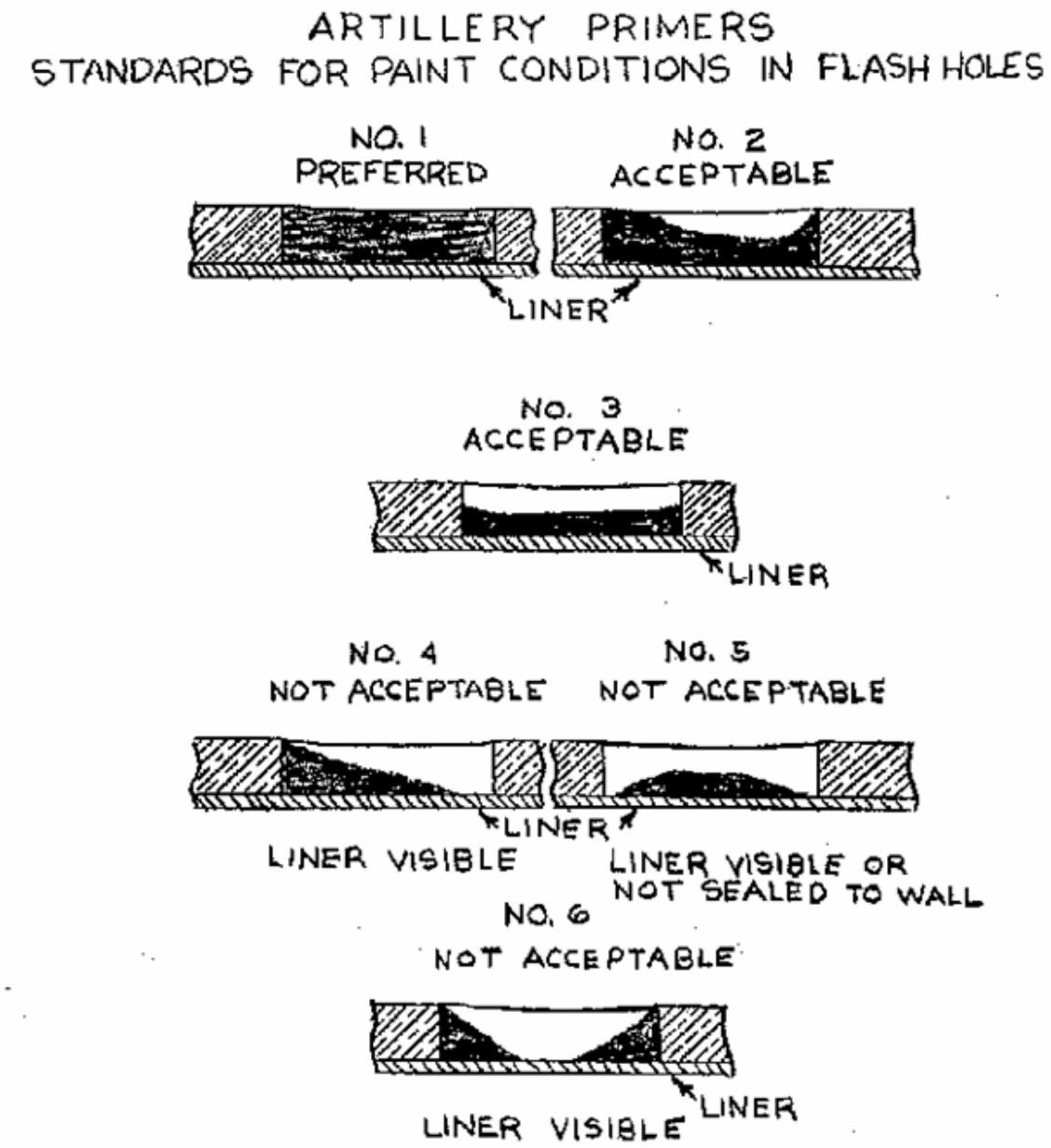


Figure I. Artillery primer standard for paint conditions in flash holes

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4.4 Methods of inspection.

4.4.1 Head assembly non-functioning. The loaded head, prior to assembling to the body, shall be assembled into the test fixture. A steel ball weighing a min. of 7.92 ounces shall be dropped from a height to give the required impact force (approximately 6 inches).

4.4.1.1 Non-functioning (full sample size). Beginning with the first lot produced, and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of 150 head assemblies shall be randomly selected from each lot for the test. If any sample fails to comply with the requirement, the lot shall be rejected. Primers subjected to this test shall be destroyed.

4.4.1.2 Non-functioning (reduced sample size). After three consecutive lots have met the criteria of 4.4.1.1, a sample of 50 head assemblies shall be randomly selected from each lot for this test. If one or more primers fail to comply with the applicable requirement, the lot shall be rejected. Primers subjected to this test shall be destroyed.

4.4.2 Head assembly functioning. The test shall be performed as specified in 4.4.1 except that the weight of the ball shall be 7.96 ounces maximum, and the height shall be adjusted to give the required impact force (approximately 20 inches).

4.4.2.1 Functioning (full sample size). Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more primers fail to comply with the applicable requirements, the lot shall be rejected. If one or two defects are found, a second sample of five hundred primers shall be tested. If the combined number of defective in both the first and second sample is three or more, the lot shall be rejected.

4.4.2.2 Functioning (reduced sample size). After three consecutive lots have met the criteria of 4.4.2.1, a sample of 125 primers shall be randomly selected from each lot for this test. If two or more primers fail to comply with the applicable requirements, the lot shall be rejected. If only one defect is found, a second sample of 250 primers shall be tested. If the combined number of defectives in both the first and second sample is two or more, the lot shall be rejected.

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4.4.3 Primer assembly functioning and pressure resistance. The sample primers shall be sent to a Government Proving Ground for testing. The primers shall be assembled to the appropriate cartridge case with M1 projectile and fired from an M119 howitzer using the PXR200-1 propelling charge to obtain the required pressure. The rounds shall be fired in an acceptable gun, based upon current gun tube condemnation criteria. The test rounds (charges and projectiles) shall be uniformly temperature conditioned at 70 degrees F ± 2.50 for a minimum of 24 hours prior to testing. The M1 projectile metal parts assembly shall be inert loaded and dummy fuzed to an as fired weight of 33.0 ± 0.2 pounds.

4.4.3.1 Primer assembly functioning and pressure resistance. (See 3.6) A sample of 25 primers shall be randomly selected from each lot for this test. If any primer fails to comply with the applicable requirement, the lot shall be rejected.

4.4.4 Determination of moisture content of black powder. The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.4.4.1 Moisture content of black powder. (See 3.4) A sample of one primer body, prior to assembly to the head, shall be randomly selected from each shifts production. Any failure to comply with the applicable requirement shall be cause for rejection of the shifts production represented by the sample.

4.4.5 Check test for deterioration of primer, percussion (M61 assembly). The check test of the M61 primer lots shall be performed as specified in MIL-P-20449 immediately before the M61 primer lot is assembled into the M28A2 primer (see 6.5 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the contract or order shall specify packaging requirements (see 6.2 and 6.3). When DOD personnel perform material packaging, those personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. The Inventory Control Point packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command, maintains packaging requirements. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components and assemblies covered by this specification are intended for use on all M28A2 Percussion Primers. The M28A2 percussion primer covered by this specification is military unique because it is used for 105MM howitzer cartridge ammunition only and has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Packaging requirements (see 5 and 6.3).

c. Requirements for submission of first article sample.

d. Requirement for submission of acceptance inspection equipment

e. Certificate of Conformance for each lot or shipment of product

e. Applicable stock number.

f. Sampling requirements for check test for deterioration of primer, percussion M61 assembly (see 6.5).

6.3 Packaging drawings. Drawings listed in this section are for reference only. The government's current packaging for the M28A2 Percussion Primers as defined by Technical Data Package (TDP) 8858935 and 8858940, is included for reference. The contractor may use the information contained in the TDP as guidance for its own design. If the contractor chooses to build to the current approved TDP (8858935 and 8858940) the Government will waive the testing required to meet the First Article performance test requirements of the packaging specification, except for Performance Oriented Packaging (POP) testing which is still required to transport hazardous materials in accordance with Code of Federal Regulations (CFR), Title 49. Unless otherwise noted, the custodian of the following drawings is the U.S. Army Tank-Automated Command- Armament Research, Development and Engineering Center, Picatinny Arsenal, NJ 07806-5000;

a. 8858935 Box, Packing, Ammunition, for Primers M28A2, M28B2 or M60A1

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b. 8858940 Box, Set-up, Packing, Ammunition for primers

6.4 Visual standards. Visual standards for evaluation of defects will be established for each contract after the first month's production. However, establishment of visual standards will not waive the requirements for compliance with the requirements of the applicable drawing.

6.5 Check test for possible deterioration of M61 percussion artillery primers. The following tests are designed to prevent the loading of deteriorated artillery primers into a major ammunition item. The test should be performed on artillery primers from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that a check test be performed, the contracting officer should insert a provision for the performance of the check tests in the contract.

6.5.1 Head assembly.

6.5.1.1 Non-functioning (see 3.6.1). The sampling plan for the test should be in accordance with table II. Any head assembly which fails to comply with the applicable requirement should be classed defective. The test should be performed as specified in 4.4.1, except that the flash tube and the head assembly should be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable. All head assemblies so tested should be scrapped.

TABLE III. Sampling plan

Sample	Sample Size	Combined Sample Size	Acceptance Number	Rejection Number
First	20	20	*	2
Second	20	40	*	2
Third	20	60	0	2
Fourth	20	80	1	3
Fifth	20	100	1	3
Sixth	20	120	1	3
Seventh	20	140	2	3

* No acceptance on this sample

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6.5.1.2 Functioning (see 3.6.2). The sampling plan for the test should be in accordance with table II. Any head assembly which fails to comply with the applicable requirement should be classed defective. The test should be performed as specified in 4.4.2, except that the flash tube and the head assembly should be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable. All head assemblies so tested should be scrapped.

TABLE IV. Sampling plan

Sample	Sample Size	Combined Sample Size	Acceptance Number	Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

* No acceptance on this sample

6.5.2 Moisture content of black powder. A sample of 25 primers should be selected for this test. The black powder should be removed from the flash tube by any suitable means. The moisture content of the powder should be determined in accordance with the method specified in Specification MIL-P-223. If the moisture content of any of the powder tested exceeds 0.7 percent, the lot of primers should be rejected. Since black powder is hygroscopic, it is advisable to place the powder in a dry, airtight container immediately upon removal, and to run the determination as soon as possible thereafter.

6.6 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test should be conducted at Government expense without cost to the contractor who loaded the M61 primer or the contractor loading and assembling the M28A2 primer. The check test should not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the primers were under the jurisdiction of either contractor.

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6.7 Critical classification notes.

6.7.1 Flash holes missing. Flash holes missing may cause improper functioning of the primer. The primer may become detached from the cartridge case or detonate, leaving material debris in the gun tube. Flash holes missing may also result in improper initiation of the propellant resulting in high or low pressure. This is likely to cause injury or death to troops firing the howitzer or cause a short round which may result in danger to ground troops.

6.7.2 Tube cracked or split. A cracked or split tube may cause improper functioning of the primer. The primer may detach from the cartridge case leaving debris in the gun tube. This may also result in improper initiation of the propellant resulting in high or low pressure resulting in injury or death to troops firing the howitzer or cause a short round which may result in danger to ground troops.

6.7.3 Firing plug or battery cup above flush. A firing plug or battery cup above flush may result in unintended initiation of the cartridge resulting in injury or death to troops firing the howitzer.

6.8 Definitions

Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard used to assure conformance of material to the contract requirements.

6.9 Supersession data. MIL-DTL-32260 supersedes the following specifications:

a. MIL-P-46562A (MU)
   3 April 1964
b. MIL-P-46565C (MU)
   8 May 1973
c. DTL8838129
   18 MAY 1998

6.10 Superseding document. DTL 8838129 specification is a ARDEC unique specification that is not in ASSIST. If a copy is required it may be requested online at Drawing-request@pica.army.mil mailto:Drawing-request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000.

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6.11 Subject term (key word) listing.

Ammunition
Artillery
Cartridge
First Article
Primer

Custodian
Army-AR

Preparing activity
Army-AR
(Project 1390-2007-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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